

## TYNSET-HANSMOEN, Norway

WMO: 012650

Lat: 62.269N

Lon: 10.732E

Elev: 482

StdP: 95.66

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -31.9      | -28.3 | -34.9                         | 0.1 | -31.8 | -31.1 | 0.2 | -28.3 | 8.6                   | -3.2 | 7.8 | -3.5 | 0.2                   | 200  | 0.688 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.0                   | 24.6            | 15.0 | 22.4 | 14.3 | 20.2 | 13.5 | 16.4                | 21.5 | 15.4 | 20.2 | 14.4 | 19.0 | 2.9                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 14.4                            | 10.8 | 18.0 | 13.3 | 10.1 | 17.2 | 12.3 | 9.5 | 16.3 | 47.4          | 21.4 | 44.7 | 20.2 | 41.9 | 18.9 | 19.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 7.2               | 6.2  | 5.4 | -34.6                      | 27.1 | 3.8                | 2.2 | -37.3                                              | 28.7 | -39.5      | 30.0 | -41.7      | 31.2 | -44.4      | 32.8 |  |  |
|                   |      |     | -34.8                      | 18.3 | 3.9                | 0.8 | -37.6                                              | 18.8 | -39.9      | 19.3 | -42.1      | 19.7 | -45.0      | 20.3 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 1.5    | -10.5 | -8.8  | -4.5  | 1.5   | 6.4   | 10.7  | 13.5  | 12.0  | 8.0   | 1.8   | -3.9  | -9.0 |
|                                                                                  | DBStd       | 9.89   | 8.15  | 7.21  | 6.27  | 3.67  | 4.01  | 3.36  | 2.99  | 2.81  | 3.27  | 4.18  | 6.05  | 8.12 |
|                                                                                  | HDD10.0     | 3379   | 636   | 526   | 451   | 254   | 126   | 32    | 5     | 12    | 76    | 256   | 417   | 588  |
|                                                                                  | HDD18.3     | 6151   | 894   | 759   | 709   | 504   | 369   | 231   | 152   | 198   | 309   | 513   | 667   | 847  |
|                                                                                  | CDD10.0     | 270    | 0     | 0     | 0     | 0     | 16    | 51    | 113   | 73    | 17    | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 3      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 1     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 113    | 0     | 0     | 0     | 0     | 10    | 28    | 69    | 6     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 12     | 0     | 0     | 0     | 0     | 1     | 2     | 9     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.0    | 1.9   | 2.0   | 2.3   | 2.3   | 2.4   | 2.3   | 2.0   | 1.8   | 2.0   | 1.9   | 1.7   | 1.7  |
| Precipitation                                                                    | PrecAvg     | 621    | 41    | 36    | 36    | 30    | 46    | 74    | 86    | 83    | 56    | 48    | 42    | 43   |
|                                                                                  | PrecMax     | 802    | 99    | 66    | 70    | 76    | 85    | 132   | 133   | 149   | 117   | 93    | 109   | 83   |
|                                                                                  | PrecMin     | 492    | 3     | 4     | 11    | 11    | 18    | 20    | 38    | 25    | 23    | 18    | 9     | 22   |
|                                                                                  | PrecStd     | 62     | 23    | 16    | 16    | 13    | 16    | 27    | 28    | 29    | 23    | 16    | 25    | 17   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 6.0   | 6.8   | 9.8   | 16.5  | 24.8  | 26.1  | 28.0  | 24.2  | 19.7  | 15.4  | 10.6  | 6.1  |
|                                                                                  |             | MCWB   | 3.1   | 2.9   | 5.0   | 8.8   | 13.2  | 14.6  | 17.4  | 15.9  | 13.2  | 12.2  | 7.3   | 3.6  |
|                                                                                  | 2%          | DB     | 3.5   | 4.4   | 7.3   | 13.4  | 21.0  | 23.5  | 25.3  | 21.8  | 17.5  | 11.6  | 7.2   | 4.5  |
|                                                                                  |             | MCWB   | 1.5   | 1.9   | 3.5   | 7.1   | 12.1  | 13.9  | 16.0  | 14.4  | 12.4  | 8.4   | 5.5   | 2.3  |
|                                                                                  | 5%          | DB     | 1.5   | 2.4   | 5.2   | 10.8  | 18.1  | 20.7  | 23.0  | 19.9  | 15.7  | 9.6   | 5.2   | 2.9  |
|                                                                                  |             | MCWB   | 0.1   | 0.5   | 1.8   | 5.5   | 10.8  | 12.5  | 15.2  | 14.1  | 11.6  | 7.2   | 3.7   | 1.1  |
|                                                                                  | 10%         | DB     | 0.1   | 0.7   | 3.6   | 8.5   | 14.7  | 18.1  | 20.6  | 17.8  | 14.0  | 8.0   | 3.2   | 1.4  |
|                                                                                  |             | MCWB   | -0.9  | -0.7  | 1.0   | 4.4   | 9.2   | 11.9  | 14.1  | 13.4  | 10.7  | 6.2   | 1.9   | 0.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.1   | 3.5   | 5.2   | 9.4   | 14.6  | 16.0  | 18.3  | 16.9  | 14.5  | 12.4  | 7.9   | 4.1  |
|                                                                                  |             | MCDB   | 5.3   | 6.1   | 9.0   | 15.2  | 21.7  | 21.7  | 25.7  | 21.5  | 18.1  | 14.8  | 10.3  | 6.4  |
|                                                                                  | 2%          | WB     | 1.7   | 2.0   | 3.7   | 7.6   | 12.9  | 14.8  | 17.0  | 15.8  | 13.1  | 8.9   | 5.7   | 2.4  |
|                                                                                  |             | MCDB   | 3.4   | 4.2   | 7.0   | 12.3  | 19.5  | 21.4  | 22.8  | 19.8  | 16.3  | 10.9  | 7.5   | 4.1  |
|                                                                                  | 5%          | WB     | 0.2   | 0.5   | 2.3   | 6.0   | 11.3  | 13.7  | 16.0  | 15.0  | 12.0  | 7.6   | 3.8   | 1.2  |
|                                                                                  |             | MCDB   | 1.3   | 2.3   | 4.6   | 10.2  | 16.9  | 19.6  | 21.3  | 18.5  | 15.0  | 9.3   | 5.1   | 2.6  |
|                                                                                  | 10%         | WB     | -0.9  | -0.6  | 1.1   | 4.5   | 9.7   | 12.5  | 14.8  | 14.0  | 10.9  | 6.2   | 2.0   | 0.0  |
|                                                                                  |             | MCDB   | 0.2   | 0.7   | 3.3   | 8.2   | 14.7  | 17.2  | 19.7  | 17.3  | 13.5  | 8.0   | 3.1   | 1.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.0   | 9.8   | 12.1  | 10.8  | 11.1  | 11.2  | 11.0  | 10.5  | 10.0  | 7.8   | 6.6   | 8.6  |
|                                                                                  |             | MCDBR  | 8.5   | 9.5   | 11.5  | 15.2  | 17.8  | 17.2  | 16.3  | 14.2  | 12.9  | 9.9   | 7.8   | 9.8  |
|                                                                                  |             | MCWBR  | 7.1   | 7.4   | 8.1   | 9.4   | 9.5   | 8.6   | 8.1   | 7.8   | 8.2   | 7.3   | 6.2   | 8.2  |
|                                                                                  | 5% WB       | MCDBR  | 8.3   | 9.0   | 10.6  | 14.0  | 15.6  | 15.0  | 14.4  | 11.6  | 11.3  | 8.8   | 7.4   | 9.5  |
|                                                                                  |             | MCWBR  | 7.0   | 7.2   | 7.6   | 9.0   | 8.5   | 7.8   | 7.6   | 6.9   | 7.4   | 6.7   | 6.0   | 8.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.198  | 0.251 | 0.283 | 0.313 | 0.327 | 0.325 | 0.338 | 0.328 | 0.297 | 0.279 | 0.210 | 0.170 |      |
|                                                                                  | taud        | 2.085  | 2.260 | 2.251 | 2.281 | 2.433 | 2.493 | 2.469 | 2.488 | 2.555 | 2.472 | 2.146 | 1.975 |      |
|                                                                                  | Ebn at Noon | 517    | 732   | 831   | 875   | 893   | 901   | 877   | 857   | 828   | 702   | 487   | 342   |      |
|                                                                                  | Edh at Noon | 32     | 59    | 87    | 105   | 99    | 95    | 95    | 86    | 66    | 49    | 29    | 18    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.25   | 0.91  | 2.44  | 4.28  | 4.88  | 4.98  | 4.64  | 3.58  | 2.27  | 1.01  | 0.32  | 0.13  |      |
|                                                                                  | RadStd      | 0.03   | 0.12  | 0.20  | 0.30  | 0.32  | 0.44  | 0.51  | 0.34  | 0.23  | 0.14  | 0.05  | 0.01  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (54 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -120        | -144    | N/A     | N/A     |

Nomenclature: See separate page